

PROGRAM 1

/* Write a Java Program to define a class, describe its constructor, overload the Constructors and instantiate its object */

```
import java.lang.*;

class student
{
    String name;
    int regno;
    int marks1,marks2,marks3;
    // null constructor
    student()
    {
        name="raju";
        regno=12345;
        marks1=56;
        marks2=47;
        marks3=78;
    }
    // parameterized constructor
    student(String n,int r,int m1,int m2,int m3)
    {
        name=n;
        regno=r;
        marks1=m1;
        marks2=m2;
        marks3=m3;
    }
    // copy constructor
    student(student s)
    {
        name=s.name;
        regno=s.regno;
        marks1=s.marks1;
        marks2=s.marks2;
        marks3=s.marks3;
    }
    void display()
    {
        System.out.println(name + "\t" +regno+ "\t" +marks1+ "\t" +marks2+ "\t" + marks3);
    }
}

class studentdemo
{
    public static void main(String arg[])
    {
        student s1=new student();
        student s2=new student("john",34266,58,96,84);
```

```
        student s3=new student(s1);
        s1.display();
        s2.display();
        s3.display();
    }
}
```

PROGRAM 2

Write a Java Program to define a class, define instance methods for setting and Retrieving values of instance variables and instantiate its object.

```
import java.lang.*;

class emp
{
    String name;
    int id;
    String address;
    void getdata(String name,int id,String address)
    {
        this.name=name;
        this.id=id;
        this.address=address;
    }
    void putdata()
    {
        System.out.println("Employee details are :");
        System.out.println("Name :" +name);
        System.out.println("ID  :" +id);
        System.out.println("Address :" +address);
    }
}

class empdemo
{
    public static void main(String arg[])
    {
        emp e=new emp();
        e.getdata("smith",76859,"gulbarga");
        e.putdata();
    }
}
```

PROGRAM 3

Write a Java Program to define a class, define instance methods and overload them and use them for dynamic method invocation.

```
import java.lang.*;

class add
{
    void display(int a,int b)
    {
        int c=a+b;
        System.out.println("The sum of " + a + " & " + b + " is " + c);
    }

    void display(double a,double b)
    { double c=a+b;
      System.out.println("The sum of " + a + " & " + b + " is " + c);
    }
}

class add_demo
{
    public static void main(String arg[])
    {
        add obj=new add();
        obj.display(10,20);
        obj.display(10.2,20.2);
    }
}
```

PROGRAM 4

/Write a Java Program to demonstrate use of sub class

```
import java.lang.*;
class parent
{
    int m;
    void get_m(int m)
    {
        this.m=m;
    }
    void display_m()
    {
        System.out.println("This is from parent : m = " +m);
    }
}
class child extends parent
{
    int n;
```

```
void get_n(int n)
{
    this.n=n;
}
void display_n()
{
    System.out.println("This is from child : n = " +n);
}
}

class childdemo
{
    public static void main(String arg[])
    {
        child c=new child();
        c.get_m(10);
        c.get_n(20);
        c.display_m();
        c.display_n();
    }
}
```

PROGRAM 5

Write a Java Program to demonstrate use of nested class.

```
import java.lang.*;
class outer
{
    int m=10;
    class inner
    {
        int n=20;
        void display()
        {
            System.out.println("m = "+m);
            System.out.println("n = "+n);
        }
    }
}

class nesteddemo
{
    public static void main(String arg[])
    {
        outer outobj=new outer();
        outer.inner inobj=outobj.new inner();
        inobj.display();
    }
}
```

PROGRAM 6

Write a Java Program to implement array of objects.

```
import java.lang.*;

public class EmployeeTest
{
    public static void main(String[] args)
    {
        Employee[] staff = new Employee[3];
        staff[0] = new Employee("Harry Hacker", 3500);
        staff[1] = new Employee("Carl Cracker", 7500);
        staff[2] = new Employee("Tony Tester", 3800);
        for (int i = 0; i < 3; i++)
            staff[i].print();
    }
}

class Employee
{
    private String name;
    private double salary;
    public Employee(String n, double s)
    {
        name = n;
        salary = s;
    }
    public void print()
    {
        System.out.println(name + " " + salary);
    }
}
```

PROGRAM 7 (A)

Write a Java program to practice using String class and its methods.

```
import java.lang.String;
class stringdemo
{
    public static void main(String arg[])
    {
        String s1=new String("gpt gulbarga");
        String s2="GPT GULBARGA";
        System.out.println(" The string s1 is : " +s1);
        System.out.println(" The string s1 is : " +s2);
        System.out.println(" Length of the string s1 is : " +s1.length());
        System.out.println(" The first accurence of r is at the position : " +s1.indexOf('r'));
        System.out.println(" The String in Upper Case : " +s1.toUpperCase());
        System.out.println(" The String in Lower Case : " +s1.toLowerCase());
    }
}
```

BASIC JAVA PROGRAMMING

```
System.out.println(" s1 equals to s2 : " +s1.equals(s2));
System.out.println(" s1 equals ignore case to s2 : " +s1.equalsIgnoreCase(s2));
int result=s1.compareTo(s2);
System.out.println("After compareTo()");
if(result==0)
    System.out.println( s1 + " is equal to "+s2);
else if(result>0)
    System.out.println( s1 + " is greather than to "+s2);
else
    System.out.println( s1 + " is smaller than to "+s2);
System.out.println(" Character at an index of 6 is :"+s1.charAt(6));
String s3=s1.substring(4,12);
System.out.println(" Extracted substring is :"+s3);
System.out.println(" After Replacing g with a in s1 : " + s1.replace('g','a'));
String s4=" This is a book ";
System.out.println(" The string s4 is :"+s4);
System.out.println(" After trim() :"+s4.trim());
}
}
```

PROGRAM 7 (B)

Write a Java program to practice using String Buffer class and its methods.

```
import java.lang.String;
class stringbufferdemo
{
    public static void main(String arg[])
    {
        StringBuffer sb=new StringBuffer("This is my college");
        System.out.println("This string sb is : " +sb);
        System.out.println("The length of the string sb is : " +sb.length());
        System.out.println("The capacity of the string sb is : " +sb.capacity());
        System.out.println("The character at an index of 6 is : " +sb.charAt(6));
        sb.setCharAt(3,'x');
        System.out.println("After setting char x at position 3 : " +sb);
        System.out.println("After appending : " +sb.append(" in gulbarga "));
        System.out.println("After inserting : " +sb.insert(19,"gpt "));
        System.out.println("After deleting : " +sb.delete(19,22));
    }
}
```

PROGRAM 8

Write a Java Program to implement Vector class and its methods.

```
import java.lang.*;
import java.util.Vector;
import java.util.Enumeration;
class vectordemo
{
    public static void main(String arg[])
```

```
{
    Vector v=new Vector();
    v.addElement("one");
    v.addElement("two");
    v.addElement("three");
    v.insertElementAt("zero",0);
    v.insertElementAt("oops",3);
    v.insertElementAt("four",5);
    System.out.println("Vector Size :"+v.size());
    System.out.println("Vector apacity :"+v.capacity());
    System.out.println(" The elements of a vector are :");
    Enumeration e=v.elements();
    while(e.hasMoreElements())
    System.out.println(e.nextElement() + " ");
    System.out.println();
    System.out.println("The first element is : " +v.firstElement());
    System.out.println("The last element is : " +v.lastElement());
    System.out.println("The object oops is found at position : "+v.indexOf("oops"));
    v.removeElement("oops");
    v.removeElementAt(1);
    System.out.println("After removing 2 elements ");
    System.out.println("Vector Size :"+v.size());
    System.out.println("The elements of vector are :");
    for(int i=0;i<v.size();i++)
    System.out.println(v.elementAt(i)+" ");
}
}
```

PROGRAM 9

Write a Java Program to implement Wrapper classes and their methods.

```
import java.io.*;
class wrapperdemo
{
    public static void main(String args[])
    {
        Float P=new Float(0);
        Float I=new Float(0);
        int y=0;
        try
        {
            DataInputStream ds=new DataInputStream(System.in);
            System.out.println("ENTER THE PRINCIPAL AMOUNT");
            System.out.flush();
            String sp=ds.readLine();
            P=Float.valueOf(sp);
            System.out.println("ENTER THE INTEREST RATE");
            System.out.flush();
            String SI=ds.readLine();
            I=Float.valueOf(SI);
            System.out.println("ENTER THE NUMBER OF YEARS");
            System.out.flush();
```

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```
        String sy=ds.readLine();
        y=Integer.parseInt(sy);
    }
    catch(Exception e)
    {
        System.out.println("INPUT OUTPUT ERROR");
        System.exit(1);
    }
    float value=loan(P.floatValue(),I.floatValue(),y);

    System.out.println("FINAL VALUE IS:"+value);

}
static float loan(float P,float I,int y)
{
    int year=1;
    float sum=P;
    while(year<=y)
    {
        sum=sum+(P*I)/100;
        year++;
    }
    return sum;
}
}
```

PROGRAM 10

Write a Java Program to implement inheritance and demonstrate use of method overriding.

```
import java.lang.*;
class A
{
    void display()
    {
        System.out.println("This is from class A ");
    }
}
class B extends A
{
    void display()
    {
        System.out.println("This is from class B ");
    }
}

class AB
{
    public static void main(String arg[])
    {
        B obj=new B();
    }
}
```



```
    obj.display();
}
}
```

PROGRAM 11

Write a Java Program to implement multilevel inheritance by applying various access controls to its data members and methods.

```
import java.io.DataInputStream;
class Student
{
    private int rollno;
    private String name;
    DataInputStream dis=new DataInputStream(System.in);
    public void getrollno()
    {
        try
        {
            System.out.println("Enter rollno ");
            rollno=Integer.parseInt(dis.readLine());
            System.out.println("Enter name ");
            name=dis.readLine();
        }
        catch(Exception e){ }
    }
    void putrollno()
    {
        System.out.println("Roll No =" +rollno);
        System.out.println("Name =" +name);
    }
}

class Marks extends Student
{
    protected int m1,m2,m3;
    void getmarks()
    {
        try
        {
            System.out.println("Enter marks :");
            m1=Integer.parseInt(dis.readLine());
            m2=Integer.parseInt(dis.readLine());
            m3=Integer.parseInt(dis.readLine());
        }
    }
}
```

```
        catch(Exception e) { }
    }
    void putmarks()
    {
        System.out.println("m1="+m1);
        System.out.println("m2="+m2);
        System.out.println("m3="+m3);
    }
}

class Result extends Marks
{
    private float total;
    void compute_display()
    {
        total=m1+m2+m3;
        System.out.println("Total marks :"+total);
    }
}

class MultilevelDemo
{
    public static void main(String arg[])
    {
        Result r=new Result();
        r.getrollno();
        r.getmarks();
        r.putrollno();
        r.putmarks();
        r.compute_display();
    }
}
```

PROGRAM 12 (A)

Write a program to demonstrate use of implementing interfaces.

```
import java.lang.*;
interface Area
{
    final static float pi=3.14F;
    float compute(float x,float y);
}

class rectangle implements Area
{
    public float compute(float x,float y)
    {
        return(pi*x*y);
    }
}

class circle implements Area
{
    public float compute(float x,float x)
```

```
        {
            return(pi*x*x);
        }
    }
}
class interfacedemo
{
    public static void main(String a[])
    {
        rectangle rect=new rectangle();
        circle cir=new circle();
        Area A;
        A=rect;
        System.out.println("Area of rectangle="+A.compute(10,20));
        A=cir;
        System.out.println("Area of circle="+A.compute(30,0));
    }
}
```

PROGRAM 12 (B)

Write a program to demonstrate use of extending interfaces.

```
import java.lang.*;
interface Area
{
    final static float pi=3.14F;
    double compute(double x,double y);
}

interface display extends Area
{
    void display_result(double result);
}

class rectangle implements display
{
    public double compute(double x,double y)
    {
        return(pi*x*y);
    }
    public void display_result(double result)
    {
        System.out.println("The Area is :"+result);
    }
}

class InterfaceExtendsDemo
{
    public static void main(String a[])
    {
        rectangle rect=new rectangle();
```

```
        double result=rect.compute(10.2,12.3);
        rect.display_result(result);
    }
}
```

PROGRAM 13

Write a Java program to implement the concept of importing classes from user defined package and creating packages.

```
package p1;
public class Student
{
    int regno;
    String name;

    public void getdata(int r,String s)
    {
        regno=r;
        name=s;
    }
    public void putdata()
    {
        System.out.println("regno = " + regno);
        System.out.println("name = " + name);
    }
}
import p1.*;

class StudentTest
{
    public static void main(String arg[])
    {
        student s=new student();
        s.getdata(123,"xyz");
        s.putdata();
    }
}
```

PROGRAM 14 (A)

Write a program to implement the concept of threading by extending Thread Class

```
import java.lang.Thread;

class A extends Thread
{
    public void run()
    {
        System.out.println("thread A is started:");
        for(int i=1;i<=5;i++)

```

```
        {
            System.out.println("\t from thread A:i="+i);
        }
        System.out.println("exit from thread A:");
    }
}

class B extends Thread
{
    public void run()
    {
        System.out.println("thread B is started:");
        for(int j=1;j<=5;j++)
        {
            System.out.println("\t from thread B:j="+j);
        }
        System.out.println("exit from thread B:");
    }
}

class C extends Thread
{
    public void run()
    {
        System.out.println("thread C is started:");
        for(int k=1;k<=5;k++)
        {
            System.out.println("\t from thread C:k="+k);
        }
        System.out.println("exit from thread C:");
    }
}

class Threadtest
{
    public static void main(String arg[])
    {
        new A().start();
        new B().start();
    }
}
```

```
        new C().start();
    }
}
```

PROGRAM 14 (B)

Write a program to implement the concept of threading by implementing Runnable Interface

```
import java.lang.Runnable;
```

```
class X implements Runnable
```

```
{
    public void run()
    {
        for(int i=1;i<10;i++)
        {
            System.out.println("\t Thread X:"+i);
        }
        System.out.println("End of Thread X");
    }
}
```

```
class Runnabletest
```

```
{
    public static void main(String arg[])
    {
        X R=new X();
        Thread T=new Thread(R);
        T.start();
    }
}
```

PROGRAM 15 (A)

Write a program to implement the concept of Exception Handling using predefined exception.

```
import java.lang.*;

class Exception_handle
{
    public static void main(String argv[])
    {
        int a=10,b=5,c=5,x,y;
        try
        {
            x=a/(b-c);
        }
        catch(ArithmeticException e)
        {
            System.out.println("DIVISION BY ZERO");
        }
        y=a/(b+c);
        System.out.println("y="+y);
    }
}
```

PROGRAM 15 (B)

Write a program to implement the concept of Exception Handling by creating user defined exceptions.

```
import java.lang.Exception;
import java.lang.*;
import java.lang.Exception;
import java.io.DataInputStream;

class MyException extends Exception
{
    MyException(String message)
    {
        super(message);
    }
}

class userdef
{
    public static void main(String a[])
    {
        int age;
        DataInputStream ds=new DataInputStream(System.in);
        try
```

```
        {
        System.out.println("Enter the age (above 15 abd below 25) :");
        age=Integer.parseInt(ds.readLine());
        if(age<15 || age> 25)
        {
            throw new MyException("Number not in range");
        }
        System.out.println(" the number is :" +age);
    }

    catch(MyException e)
    {
        System.out.println("Caught MyException");
        System.out.println(e.getMessage());
    }
    catch(Exception e){ System.out.println(e); }

    }
}
```


*****OUTPUT 1*****

```
c:\jdk1.6.0_26\bin>java userdef
```

```
Enter the age (above 15 abd below 25) :
```

```
6
```

```
Caught MyException
```

```
Number not in range
```

